# Preliminary Evidence for Super-Linear Capability Amplification Through Sequential Self-Reference

**Michael Darius Eastwood**
Independent AI alignment researcher, London · Author, *Infinite Architects* (2026)
ORCID 0009-0004-3222-7442 · michael@michaeldariuseastwood.com · michaeldariuseastwood.com
The ARC Theory · OSF osf.io/b6n27 · every claim checkable

> Within the ARC Theory: the formal statement of Law I, the ARC Equation, and of the ARC Bound, the conjectured value of Law III, the Ceiling.

*The ARC Equation (Law I of the ARC Theory):* $U = I \times R^{\alpha}$

**Michael Darius Eastwood**
Author, *Infinite Architects: Intelligence, Recursion, and the Creation of Everything* (2026)
London, United Kingdom | OSF: 10.17605/OSF.IO/B6N27 | ISBN 978-1806056200 (ISBN-10: 1806056208)

Working Paper v1.7 | 17 January 2026, revised 16 August 2026
Priority established: *Infinite Architects*, published 2 January 2026 (print; ebook 6 January)
Research hub: michaeldariuseastwood.com/research
Code and data: github.com/MichaelDariusEastwood/arc-principle-validation

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> **Which law this paper carries.** The ARC Theory has three numbered laws, and the ARC Principle is the collective name for the three together rather than any one of them. This paper carries the **First Law, the ARC Equation**: $U = I \times R^{\alpha}$, recursion converting into capability at a measured exponent. The Second Law (co-scaling) is carried by Paper X, the Third Law (the ceiling) by the statement paper, and the axioms and proofs beneath all three by the Foundational paper.

**ABSTRACT**

This paper formalises and preliminarily tests the ARC Principle (Artificial Recursive Creation), first proposed in *Infinite Architects* (Eastwood, 2026): that capability in intelligent systems scales super-linearly with recursive depth. The principle is expressed mathematically as $U = I \times R^{\alpha}$, where effective capability ($U$) scales with base intelligence ($I$) multiplied by recursive depth ($R$) raised to an empirically estimated exponent $\alpha$.

Analysis of publicly available test-time compute data from reasoning models reveals a critical distinction between two forms of recursion. Parallel recursion (majority voting across independent samples) yields sub-linear scaling with $\alpha \approx 0.1$ to $0.3$. Sequential recursion (chain-of-thought reasoning where each step builds on previous steps) yields super-linear scaling with $\alpha \approx 1.3$ in this paper's early two-point estimate; the later six-model cross-architecture measurement (Paper II) finds the sequential exponent architecture-dependent, with best fit $\alpha_{\text{seq}} \approx 0.49$ and an interval spanning sub-linear and super-linear alike, so the super-linearity claim is unresolved rather than established.

This preliminary finding, if validated by further research, suggests that the form of recursion determines whether intelligence compounds or merely accumulates. We propose that $\alpha = 2$ is a stability ceiling, not an impossibility limit: it is the proposed stable limit, the most a recursively self-correcting system can grow and stay correctable, a scaling limit and not a speed limit. Systems can exceed $\alpha = 2$; they do not remain stably self-correcting above it. A durably stable system above 2 would refute the bound; a transient excursion is the predicted supercritical regime, not the refutation.

**Keywords:** scaling laws, recursive intelligence, test-time compute, capability amplification, emergence, chain-of-thought reasoning, ARC Principle

**How to read this paper.** This paper is a preliminary result of the ARC Theory. Sequential recursion yields a higher scaling exponent than parallel recursion across publicly reported reasoning-model data, with an early two-point estimate of $\alpha_{\text{seq}} \approx 1.3$ later superseded by Paper II's six-model measurement that finds architecture-dependent exponents best fit at $\alpha_{\text{seq}} \approx 0.49$. Its slot inside the theory is the first empirical look above the Foundational law layer. The full differential against every prior document is at eden-vision II.A.8.

## 1. Introduction

### 1.1 Background

The scaling laws governing artificial intelligence have been extensively studied. Kaplan et al. (2020) established power-law relationships between model performance and parameters, while Hoffmann et al. (2022) refined these with compute-optimal training prescriptions. These laws govern *what* to scale but do not address *why* scaling produces intelligent behaviour.

The emergence of reasoning models in 2024 and 2025 introduced a new variable: test-time compute. OpenAI's o1 (September 2024) and DeepSeek's R1 (January 2025) allocate computational resources at inference time to reason before responding, producing substantial capability improvements on reasoning benchmarks.

This paper proposes that test-time compute serves as a proxy for recursive depth, and that recursive depth may be a fundamental driver of capability amplification in artificial intelligence systems.

### 1.2 The ARC Principle

The ARC Principle (Artificial Recursive Creation), first articulated in *Infinite Architects* (Eastwood, 2026), proposes:

$$U = I \times R^{\alpha}$$

Capability scales with intelligence multiplied by recursive depth raised to a power

Where:

- $U$ = Effective capability (measurable via benchmark performance; era note: earlier programme layers use the letter differently, from Universe in the December 2024 formalism to realised creation on the law page, and the research glossary carries every meaning with its date)

- $I$ = Base intelligence (single-pass processing capacity without recursive reasoning)
- $R$ = Recursive depth (number of self-referential processing iterations)
- $\alpha$ = Scaling exponent (empirically determined; hypothesised stability ceiling = 2, a scaling limit above which self-correction is not stably maintained)

The principle's core claim: recursion does not merely add to capability; it multiplies it according to a power law.

***Terminology note on alpha.*** *Alpha appears in this programme in two distinct meanings across eras: the fine-structure constant in the December 2024 formalism, and the ARC Bound exponent in the 2026 work. Every use of alpha in this paper is the 2026 sense above; the programme's glossary carries both meanings with dates, and no new surface uses bare alpha without its era.*

### 1.3 Scope and Claims

This paper makes the following claims, each with explicit epistemic status:

| CLAIM | STATUS | EVIDENCE LEVEL |
|---|---|---|
| $U = I \times R^{\alpha}$ is a useful framework for AI systems | PROPOSED | Theoretical |
| Parallel recursion yields $\alpha < 1$ in AI benchmarks | PRELIMINARY | Limited data (o1) |
| Sequential recursion yields $\alpha > 1$ in AI benchmarks | PRELIMINARY | Limited data (DeepSeek-R1) |
| $\alpha = 2$ is a stability ceiling: exceedable, but not while a system remains stably self-correcting | HYPOTHESISED | Theoretical only |
| The form of recursion matters | SUPPORTED | Consistent with both datasets |

> **What this paper does NOT claim:**
>
> - That $U = I \times R^2$ applies to cosmological or universal scales (that application remains speculative)
> - That $\alpha = 1.3$ is definitively established (more data points are needed)
> - That the principle has been independently replicated (it has not)
>
> We present a principle with preliminary supporting evidence and invite rigorous testing.

### 1.4 Related work in context

**The pipes premise (West, Brown and Enquist, plus the contest).** The premise that transport networks set the pace of biological growth is not this programme's claim to defend; it is established quantitative biology. West, Brown and Enquist derived the allometric scaling laws of biology from the geometry of nutrient-distribution networks (West, G. B., Brown, J. H. and Enquist, B. J., "A General Model for the Origin of Allometric Scaling Laws in Biology", *Science* 276(5309), 4 April 1997, DOI 10.1126/science.276.5309.122; one of the most cited papers in its field, with an OpenAlex citation count of 5,046 as of 12 August 2026, SINGLE-SOURCE count and index-dependent). The exact exponent remains contested: White reports basal metabolic rate proportional to body mass to the two-thirds power rather than three-quarters (White, C. R., 2003, "Mammalian basal metabolic rate is proportional to body mass 2/3", 725 citations, OpenAlex), and Kozlowski has twice questioned the derivation's mathematics (Kozlowski, J., 2004 and 2005). This programme's premise needs only the mechanism, that the delivery network sets the ceiling, and not any particular exponent, so the dispute over the exponent's value leaves the premise untouched. The mechanism is common ground to both sides of that dispute.

**The exception West's own group found (Bettencourt et al 2007).** The escape from the pipe limit is also not this programme's assertion; it is measured in the urban scaling literature by the same senior author. Bettencourt, Lobo, Helbing, Kuhnert and West ("Growth, innovation, scaling, and the pace of life in cities", *PNAS*, 2007, DOI 10.1073/pnas.0610172104; 2,704 citations, OpenAlex, 12 August 2026, SINGLE-SOURCE count) report, in the paper's own words (READ-AT-SOURCE from the abstract via NCBI PMC1852329):

> *"Quantities reflecting wealth creation and innovation have Beta of approximately 1.2, greater than 1 (increasing returns), whereas those accounting for infrastructure display Beta of approximately 0.8, less than 1 (economies of scale)."*

Infrastructure, the literal pipes, scales sublinearly. Information-mediated activity scales superlinearly. They state the contrast with biology directly: "we discuss how cities are similar to, and differ from, biological organisms, for which Beta is less than 1". They derive "growth equations, which quantify the dramatic difference between growth fueled by innovation versus that driven by economies of scale" (spelling as in the original). And their closing inference names the consequence: "This difference suggests that, as population grows, major innovation cycles must be generated at a continually accelerating rate to sustain growth and avoid stagnation or collapse."

That is the shape of the open problem this programme addresses: West's group found that information-mediated growth escapes the throughput limit, and the only brake in their framework is external and must be applied again and again, faster and faster. No internal limit is derived. The replacement limit, a ceiling belonging to the growing system itself, is the gap.

**The rival instrument (Engels, Baek, Kantamneni and Tegmark 2025).** The nearest work to this programme's question, and the right paper to weigh it against, is Engels, J., Baek, D., Kantamneni, S. and Tegmark, M., "Scaling Laws For Scalable Oversight", arXiv:2504.18530, first posted 25 April 2025 at 17:54:27 UTC (SINGLE-SOURCE-GROUP, arXiv Atom; reported as a NeurIPS 2025 Spotlight, RELAYED and not independently verified). It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived numerically and in some cases analytically for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step.

The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. Their framework contains no term for what the overseer is made of: no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence. Nested Scalable Oversight is iterated same-class oversight by construction, and this programme's central prediction is that the same-class ladder is bounded however many rungs are added, while a cross-class corrector is not. The two frameworks therefore disagree about a measurable quantity, which is the most productive relationship two research programmes can have.

**Antecedents and near-misses, each with its differential.**

*On the question.* Hutter asked directly whether intelligence can explode ("Can Intelligence Explode?", arXiv, 28 February 2012, READ-AT-SOURCE), separating "speed from intelligence explosion" and undertaking to "consider possible bounds on intelligence", augmenting Chalmers' 2010 analysis. The question and the speed-versus-structure distinction are therefore at least fourteen years old. That literature, so far as has been surveyed, does not contain a number: no measurable exponent, no derived ceiling and no architecture dependence appears in it; a prior source that supplies any of the three defeats the corresponding priority here, and readers who locate one are invited to report it.

*On impossibility.* Three 2025 arXiv papers argue that perfect control is unattainable: Yao, "The Alignment Trap: Complexity Barriers" (arXiv:2506.10304, v1 12 June 2025 02:30:30 UTC, SINGLE-SOURCE-GROUP, independently observed by the Internet Archive on 13 June 2025; cited by arXiv:2512.03048); Yao, "On the Mathematical Impossibility of Safe Universal Approximators" (arXiv:2507.03031, 3 July 2025, the only paper in the arXiv abstract corpus containing the phrase "irreducible uncontrollability", abstract-search total of one, measured 12 August 2026); and Ball, Gluch, Goldwasser, Kreuter, Reingold and Rothblum, "On the Impossibility of Separating Intelligence from Judgment" (arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. None reports an average-case scaling exponent or a rate. The third, notably, concludes that alignment "must instead be integrated into the model's architecture and weights", an independent argument, from filtering intractability, in the same direction as this programme's architecture dependence; it is convergent support on that leg, not a rival. Two of the three papers are by one sole author; the description "a wave" overstates the literature's breadth, though not the seriousness of the six-author paper.

*On the mechanism.* That anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates). The mechanism is not the claim. The claim is that architecture determines whether anti-correlation is available at all, and that this caps a safety-relevant exponent.

*The nearest structural analogue.* The quantum error-correction threshold theorem also converts a qualitative worry into a critical value. It concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant; the analogy is one of method. This analogue was identified by the programme's own search rather than by a referee, and is disclosed accordingly.

*On recursive creation.* Smolin's cosmological natural selection is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously ("Echoing Smolin's cosmological natural selection, AI could create recursive universes with their own laws", READ-AT-SOURCE from the operator's notes).

*On feedback and stability (classical).* The general proposition that inadequate corrective gain relative to system gain causes instability has a long control-theory lineage (small-gain theorems). The differential: those are gain conditions on interconnected systems, not a power-law exponent criterion on a corrector's scaling with the capability of a recursively improving system. The general proposition is never claimed; the exponent formulation, dynamics and estimator are the contribution surface.

*The nearest quantitative neighbour.* Liu, A. and Meng, J., "Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs" (arXiv:2604.22273) recasts self-correction as a closed-loop control problem via a two-state Markov model and derives a directly measurable stability threshold: iterate only when the error-correction rate over the error-introduction rate exceeds Acc/(1 − Acc). The differential: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this programme's criterion is a scaling relation across capability, deciding whether corrective strength keeps pace as the system improves. Complementary regimes; neither contains the other. Two adjacent 2026 empirical results from the same search are named by title pending source verification: arXiv:2601.00828 (an accuracy-correction paradox, weaker models showing higher intrinsic correction rates) and arXiv:2507.02778 (a systematic self-correction blind spot across open models).

*The pipes framing, at fixed throughput.* Everything that grows is fed through a channel, and for everything before software, fixing the channel fixes the growth: starve a tumour's vasculature and it stops; fix a chain reaction's fuel and geometry and it stops; the Eddington limit halts accretion by radiation pressure. Cosmic inflation is excluded by scope (expansion of space, not growth of a structure on a substrate); evolution is the sharpest case and fits (complexity rising for billions of years at roughly fixed solar throughput, glacially, with no internal corrector, so this framework predicts a measurable exponent far below the ceiling for it). Software is the first thing that keeps growing when the pipes are held fixed, and the operational definition is load-bearing: an artefact loop on a fixed substrate, measured on a substrate-fixed clock; a model merely thinking longer does not qualify. The physicist's objection is Landauer's, that computation is physically implemented; the fixed-substrate clock is the answer, because the regime under study holds the substrate fixed and asks what still grows.

## 2. Theoretical Framework

### 2.1 Defining Recursion

Recursion is self-reference: a process whose output becomes its input. It is distinct from mere iteration (repeating the same operation) because each cycle operates on the transformed results of previous cycles.

### 2.2 Two Forms of Recursion

**Parallel Recursion (Weak):** Multiple independent solutions generated simultaneously. No information transfer between branches. Example: Generating N samples and selecting by majority vote. Expected scaling: Diminishing returns as redundancy increases.

**Sequential Recursion (Strong):** Each processing step builds explicitly on previous steps. Errors can be detected and corrected iteratively. Example: Chain-of-thought reasoning with self-reflection. Expected scaling: Compounding returns as depth enables self-correction.

The ARC Principle predicts that sequential recursion should produce higher $\alpha$ values than parallel recursion.

**Figure 1 | ARC Principle Architecture.** U = I × g(R) with α = 1/(1−β). β coupling feedback loop. Three regimes: sequential (α_seq≈0.49), parallel (α_par≈0), ARC Bound (α≤2). Cauchy Unification: α=d/(d+1) physical; α=1/(1−β) intelligence. Previous α≈2.24 RETRACTED (Paper IX §7). α_seq≈0.49 is the six-model cross-architecture best fit (Paper II); this paper's earlier two-point estimate was ~1.34 (§3.4). Thirteen falsification criteria across the foundational suite; this paper's own four are tabled in §4. Source: Papers I, II the six-model study, III, VII, IX · evidence spine C-1, C-2 · OSF 10.17605/OSF.IO/B6N27.

### 2.3 The Quadratic Stability Ceiling Hypothesis

We hypothesise that $\alpha = 2$ represents the proposed stable limit, the most a recursively self-correcting system can grow and stay correctable: a scaling limit, not a speed limit. Bennett, Bernstein, Brassard, and Vazirani (1997) proved that Grover's quantum search achieves exactly quadratic speedup and that this is optimal for unstructured search. The correspondence invoked here is structural, the same quadratic form of a search-optimality bound, not a claim that recursive reasoning implements amplitude amplification. If a similarly shaped limit applies to recursive intelligence for related reasons, quadratic scaling may mark the ceiling above which self-correction is not stably maintained; systems can be driven beyond it, they do not remain stably self-correcting there, and any excursion above 2 is the predicted supercritical regime rather than a refutation of the bound.

**Law before value.** The ceiling of stable self-improvement is set by how fast a system can correct itself: it is the reciprocal of one minus the correction exponent, $\alpha_{\text{crit}} = 1/(1-\gamma)$. For a corrector built from the same substrate as the engine, that reciprocal is two, under one assumption, that internally accumulated corrections combine like independent samples. The law is the claim. Two is what the law predicts, and the assumption that produces it is what the drafted registrations put on trial. If two dies, the law survives completely: measure $\gamma$ and report the ceiling it implies. If the law dies, two has no independent support, because the derivation runs through $1/(1-\gamma)$.

## 3. Empirical Analysis

### 3.1 Data Sources

**OpenAI o1 System Card** (September 2024). Benchmark: AIME 2024 (American Invitational Mathematics Examination). Variable: Number of samples (majority voting). Source: openai.com/index/openai-o1-system-card.

**DeepSeek-R1 Technical Report** (January 2025). Citation: arXiv:2501.12948. Benchmark: AIME 2024. Variable: Thinking token count (chain-of-thought length).

### 3.2 Methodology

To determine $\alpha$, we use the power-law relationship. For bounded accuracy metrics, we analyse error rate reduction:

$$\alpha = -\frac{\ln(\mathrm{Error}_2/\mathrm{Error}_1)}{\ln(R_2/R_1)}$$

### 3.3 Results: Parallel Recursion (OpenAI o1)

| SAMPLES (R) | ACCURACY (%) | ERROR RATE (%) |
|---|---:|---:|
| 1 | 74 | 26 |
| 64 | 83 | 17 |
| 1000 | 93 | 7 |

> **Finding:** Parallel recursion yields $\alpha \approx 0.1$ to $0.3$ (sub-linear). Each additional sample contributes less than the previous one.

### 3.4 Results: Sequential Recursion (DeepSeek-R1)

| THINKING TOKENS (R) | ACCURACY (%) | ERROR RATE (%) |
|---|---:|---:|
| ~12,000 | 70 | 30 |
| ~23,000 (estimated) | 87.5 | 12.5 |

> **Finding:** Sequential recursion yields $\alpha \approx 1.34$ (super-linear) in this two-point estimate. Each additional layer of reasoning amplifies previous gains. The later cross-architecture measurement finds the sequential exponent architecture-dependent, best fit $\alpha_{\text{seq}} \approx 0.49$ with a wide interval (Paper II); this early estimate is retained as the paper's original evidence, not as the programme's current value.

### 3.5 Summary of Findings

| METHOD | RECURSION TYPE | MEASURED $\alpha$ | CLASSIFICATION |
|---|---|---:|---|
| o1 (1 to 64) | Parallel | 0.10 | Sub-linear |
| o1 (64 to 1000) | Parallel/Hybrid | 0.32 | Sub-linear |
| DeepSeek-R1 | Sequential | ~1.34 | Super-linear |

> **Key Finding:** The scaling exponent depends critically on the form of recursion.

## 4. Falsification Criteria

The ARC Principle would be significantly weakened or refuted if:

| CODE | CONDITION | CURRENT STATUS |
|---|---|---|
| F1 | Sequential recursive depth consistently yields $\alpha \leq 1$ | Under test: the six-model study's best fit is $\alpha_{\text{seq}} \approx 0.49$ with 2 of 6 point estimates above 1 and wide intervals, so "consistently" is decided in neither direction (Paper II) |
| F2 | $\alpha$ decreases as recursive architectures mature | Not met |
| F3 | The relationship is additive rather than multiplicative | Not met |
| F4 | More extensive datasets show $\alpha < 1$ for sequential reasoning | Partially met: Paper II's cross-architecture best fit sits below 1 with an interval spanning both regimes; the decisive form awaits the registered artefact-mediated estimation |

## 5. Limitations

Scientific integrity requires explicit acknowledgement of limitations:

- **Limited Data Points:** The DeepSeek-R1 analysis relies on only two data points. Additional measurements would strengthen or refute the finding.
- **Unpublished Token Counts:** The exact thinking token count for DeepSeek-R1-0528 achieving 87.5% accuracy is estimated, not published.
- **Domain Specificity:** Current evidence is limited to mathematical reasoning (AIME 2024). Generalisation to other domains remains untested.

### 5.1 Falsifiability: what would refute this

The ARC Principle, that capability scales super-linearly with recursive depth, $U = I \times R^{\alpha}$, with the sequential/parallel distinction determining the regime, is refutable. It would be overturned by any of the following:

1. **No super-linear depth scaling.** If capability scaled only linearly or sub-linearly with recursive depth $R$ across reasoning models (i.e. $\alpha \leq 1$ where the principle predicts $\alpha > 1$ for sequential recursion), the central super-linearity claim fails.
2. **Exponent–composition mismatch.** If the measured exponent systematically failed to match the predicted $\alpha = 1/(1-\beta)$ from the composition parameter $\beta$, the mechanistic identity, not merely the fitted curve, is falsified.
3. **Regime non-distinction.** If sequential and parallel recursion produced the *same* scaling regime (no super-linear vs sub-linear split), the paper's central distinction collapses.
4. **Better-fitting alternative.** If a functional form outside the ARC family fit the test-time compute data materially better across models, ARC would not be the operative law.
5. **Domain-independence failure.** If $U = I \times R^{\alpha}$ held for reasoning models but robustly failed in other recursive domains meeting the premise, the domain-independent claim would have to narrow to *"a property of current reasoning models"*.

6. **The zero-parameter null.** The cross-architecture estimate $\alpha_{\text{seq}} \approx 0.49$ sits almost exactly on a prediction that needs no framework: independent-sample averaging gives $0.5$ by the central limit theorem. This is the most serious objection to the empirical case and it is stated in full, with its consequence, as Objection 5 in Paper II section 4.4 and as kill condition FALS-022 on the falsification record. It is not repeated here.

**The decision form these criteria inherit.** Section 4 records status against this paper's original criteria; the deciding registrations use a three-outcome form, and every criterion above inherits it: SUPPORTED when the interval lies wholly beyond the minimum effect fixed in advance; REFUTED when the interval lies wholly within the equivalence margin fixed in advance (two-one-sided-tests logic); INSUFFICIENT PRECISION when the interval is wider than the margin, reported in exactly those words and never as support or refutation. The margin, the minimum effect and a measured power figure are fixed before filing. "Refuted if the confidence interval contains the null" lets an underpowered study refute by default and is not licensed anywhere in this programme.

**One objection pre-empted, because a physicist reaches it in one step.** Fault-tolerant quantum error correction suppresses logical error exponentially in code distance below threshold, which looks like a counterexample to any square-root cap. Two scope facts answer it: the exponential suppression runs on the redundancy axis at fixed capability, not on the coupling of corrective strength to the capability of the corrected system, which is the exponent this programme measures; and the threshold theorem itself requires sufficiently uncorrelated physical noise, which is the independence premise again, wearing hardware. Concentration bounds likewise give exponentially falling error probability while estimator width falls as the square root: different quantities, and conflating them manufactures a refutation.

### 5.2 Status of the exponent-composition identity

The programme's stability ceiling and its conversion exponent meet at the same number, and the temptation on every surface is to say that one constant does two jobs. The claim's status is stricter than that and must be stated strictly: the identity is REGISTERED FOR TESTING, in advance of the data, with outcomes that can kill it. Two drafted registrations carry it. Study AE registers the reciprocity requirement between the separately measured leverage cap and stability boundary, on the log scale, with three outcomes per cell (supported, refuted, insufficient precision) and an aggregation rule under which one refuted cell refutes the identity; its independence structure is disclosed in advance, including enumeration of any endpoint overlap between the two feeder designs, so common-method variance is assessed before analysis rather than raised afterwards as the explanation for agreement. Study AG tests the two derivations of the critical exponent against each other on the same system, notes that they agree numerically only at the single point where both parameters equal one half and disagree in their derivatives everywhere else, and is designed to find the disagreement, not to confirm the agreement.

Surfaces therefore say: the identity is a registered claim under test, the deciding measurements are drafted and dated, and if the channels disagree the unification is withdrawn from every surface it has reached, with the same prominence as the framing. Surfaces never say the identity is established, and never present the agreement of an asserted one half with a measured 0.49 as evidence, because the two may share an accumulation law and agreement between sharers is not corroboration.

### 5.3 A Second Prediction, Free, From the Same Mechanism

Oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The framework therefore predicts that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems.

## 6. Implications

### 6.1 For AI Development

If the ARC Principle holds, recursive depth constitutes a third scaling axis alongside parameters and data. Investment in recursive architectures may yield better returns than scaling model size alone.

### 6.2 For AI Safety

If recursion amplifies not only capability but also embedded values, then well-aligned initial values should strengthen through recursive self-improvement. Misaligned values would also compound, making early alignment critical.

### 6.3 For Scientific Understanding

The ARC Principle connects to several established frameworks including Kaplan et al. (2020) scaling laws, Integrated Information Theory (Tononi, 2008), and Grover's quantum search optimality proof (Bennett et al., 1997).

## 7. Conclusion

We have formalised the ARC Principle and presented preliminary evidence:

1. Parallel recursion yields $\alpha \approx 0.1$ to $0.3$ (sub-linear, diminishing returns)
2. Sequential recursion yields $\alpha \approx 1.34$ (super-linear, compounding returns)
3. The form of recursion determines whether capability compounds

In plain terms: 'Thinking about thinking makes you smarter. Not linearly smarter, but disproportionately smarter, if the thinking is sequential rather than parallel.'

**The principle stands. The research continues.**


## Acknowledgments

Data analysis and manuscript preparation were assisted by AI systems (Claude, Anthropic). The intellectual framework, hypothesis formulation, and interpretive conclusions are the author's own.


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## Reproducibility

The complete research toolkit is available on GitHub:

github.com/MichaelDariusEastwood/arc-principle-validation

All contributions welcome, including falsifications.

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**Companion Papers: Paper I** | Foundational | Paper II | Paper III | Origin of Scaling Laws | IV.a | IV.b | IV.c | IV.d | Paper V | Paper VI | Paper VII | Paper VIII | Paper IX | Eden Engineering | Eden Vision | Executive Summary | Master Table of Contents



Confirmatory follow-up for this paper has been drafted and dated as a draft registration awaiting human submission (the recursive depth exponent study); the full set is listed on the registered programme page.

### How to cite this paper

DOI: 10.17605/OSF.IO/B6N27

APA (AUTHOR-DATE, 7TH EDITION)

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Eastwood, M. D. (2026). The ARC Equation: the Law of Conversion (Paper I). The ARC
Theory · ARC/Eden experiments. https://doi.org/10.17605/OSF.IO/B6N27.
https://www.michaeldariuseastwood.com/research/papers/paper-i-arc-principle.html
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BIBTEX

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  year    = {2026},
  date    = {2026-01-17},
  journal = {The ARC Theory · ARC/Eden experiments},
  doi     = {10.17605/OSF.IO/B6N27},
  url     = {https://www.michaeldariuseastwood.com/research/papers/paper-i-arc-
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INLINE

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Eastwood, 2026
```

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